THE UNIVERSITY OF AUCKLAND

SEMESTER ONE 2025
Campus: City

COMPUTER SCIENCE

COMPSCI 713: AI Fundamentals

Sample TEST

This is a sample test, very close in style and format to the actual test you will take in week 7. You are encouraged to discuss it with classmates and on Ed. In the actual test you will have the instructions below.

Time Allowed: 60 minutes (5 minutes reading time + 55 min answering time)

NOTES:

- This test is out of **15** marks.
- There are 6 short-answer questions. Attempt **ALL** questions.
- This is a restricted book test. You are only allowed to use a **double-sided page of handwritten notes**.
- Digital devices, including calculators and phones are not permitted.
- Write your answers directly after each question.
- Cross out notes and other work you do not wish to be assessed.
- We privilege quality over quantity, i.e., you do not need to write very long answers. Be concise and clear.
- Write as clearly as possible, in plain English.
- Overflow space is available at the end of the test paper, indicate at the end of the original question if you are using overflow space.

Question 1: [3 marks]

This question tests your understanding of two fundamental types of symbolic logic: *propositional logic* – which deals with complete statements and logical connectives; and *first-order logic* – which involves quantifiers and reasoning about individuals in a domain. Answer both part (a) and part (b).

(a) Consider the example below.

> In a secure facility, a person is granted entry only if they provide both a valid ID and a fingerprint match.
>
> Let the propositions be:
> I: The person has a valid ID
> F: The fingerprint matches
> E: The person is granted entry
>
> The rule is logically expressed as:
> $(I \land F) \rightarrow E$
>
> Today, we observed that the person was not granted entry.

Use propositional logic to deduce what must be true about P and Q. Show your steps clearly. **[1 mark]**

(b) Consider the example below in answering (i) and (ii).

A biologist claims: "Not all birds in this region can fly." Let the domain be all birds in this region, and Fly(x) means bird x can fly.

i. Write this claim in formal first-order logic. **[1 mark]**

ii. Provide a realistic example (in one sentence) that would make the statement true. **[1 mark]**

Question 2: [2 marks]

Consider the example below in answering (a) and (b).

A smart home system uses a Logic Neural Network (LNN) to determine whether to turn on the heating system based on environmental and occupancy conditions. The decision rule is:

"If it is cold and someone is at home, then the heating system should be turned on."

This rule is encoded logically as:

$$\text{HeatingOn} \leftarrow \text{Cold} \otimes \text{AtHome}$$

Here:

- $\otimes$ represents the conjunction (AND) operator implemented in LNN as a differentiable function.

(a) What does this rule represent in natural language, and how is it different from a standard Boolean rule? **[1 mark]**

(b) Suppose the network receives the following inputs:

Cold = 0.9

AtHome = 0.4

Explain how the LNN would likely compute the truth value of HeatingOn, and whether the system would activate heating or not. (You may answer qualitatively; exact numerical output is not required.) **[1 mark]**

Question 3: [2 marks]

What is the role of entity and relation embeddings in knowledge graph completion? Introduce a common Knowledge Graph Inference task and provide one example of how such embeddings help in a typical KG task.

Question 4: [2 marks]

Consider a robot soccer league where the rules allow all robots to access an overhead camera view of the field (so, they know the position of all teammates, opponents and the ball), but they cannot send signals (other than visual signals) to teammates.

Describe one strategy or collective behaviour that can be effective under these conditions.

Question 5: [3 marks]

You're trying to predict the 1-month change in value for a stock based on a set of 225 different features of the stock (aspects of the company, its products and market conditions). You're going to use a random forest with feature bagging to make the estimate.

(a) Describe how the bagging algorithm selects features for any given tree **[2 marks]**

(b) Why is feature bagging considered a good idea? **[1 mark]**

Question 6: [3 marks]

Both forward and backward chaining are useful in medical diagnosis, where a doctor decides what is likely to be wrong with a patient based on symptoms and observations. A patient might initially volunteer to the doctor one symptom, e.g. a runny nose, which could be due to the common cold. But runny nose can also be due to allergies (in which case the patient may have a history of allergies) or measles (where the patient develops a distinctive rash).

In the context of these six concepts (runny nose, history of allergies, rash, common cold, allergies and measles) describe an instance of backward chaining reasoning for a patient with a runny nose.

OVERFLOW PAGES